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Interruption method for commutation failure caused cascading reaction of HVDC with wind farm integration under grid fault

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ABSTRACT

Line-commuted converter based high voltage direct current (LCC-HVDC) is a typical mode for wind power transmission. The fault of the receiving-end grid of LCC-HVDC can easily lead to the commutation failure at the inverter station. Improper controls during the recovery process of first commutation failure may cause the subsequent commutation failure, and lead to the high voltage ride through of wind generators at the sending-end grid. The regional and inter-regional cascading reactions are produced by commutation failure. Existing researches severally focus on control of suppressing subsequent commutation failure or overvoltage. The demand for commutation failure recovery of the inverter and overvoltage suppression at the sending-end grid cannot be satisfied simultaneously, and may even aggravate the cascading process. The stability of the power system will be threatened. Therefore, the new idea of feasible power region and adaptive current control of inverter station is proposed to interrupt commutation failure caused cascading reaction. The feasible ranges of DC current to avoid subsequent commutation failure and overvoltage are derived. The feasible power region is modeled by the active power and exchanged reactive power of inverter station. The principle of interrupting commutation failure caused cascading reaction based on adaptive current control of inverter station is proposed. Then, the command value calculation and implementation method of adaptive current control are further proposed. Finally, the performance of the proposed control method and voltage-dependent current order limiter (VDCOL) control is compared based on the CIGRE LCC-HVDC standard system, and the effectiveness of theoretical research is verified by the simulation results.

1. Introduction

Driven by the global energy crisis and environmental problems, wind power has developed rapidly worldwide in recent years, and the proportion of wind power in the power grid is getting higher and higher [1]. Line commutated converter based high voltage direct current (LCC-HVDC) is appropriate for wind power transmission due to the advantages of low loss, large capacity, long distance and flexible power regulation [2,3]. Therefore, the new pattern of wind power transmitted through LCC-HVDC is emerged massively. Due to the special structure and the strong coupling of power electronic equipment, the wind power transmission system via LCC-HVDC is prone to cascading reaction under grid fault, which brings severe challenges to the safe operation of power systems.

Commutation failure (CF) of LCC-HVDC is the current focus of researchers [4]. The voltage fluctuation caused by fault at the receiving-end grid can easily produce first CF (FSF) of LCC-HVDC. It is easy to cause subsequent CF (SCF) when the control is improper. Repeated power shocks to the grid are generated under SCF, which may lead to the blocking of converter station and threaten the safe and stable operation of power systems. In addition, during the FCF, the reactive power consumption of the rectifier station fluctuates sharply with the change of the transmitted power, resulting in power frequency overvoltage of the AC bus of the rectifier station [5]. The overvoltage causes a significant increase of the access point voltage of the wind farm near the LCC-HVDC through electromagnetic coupling. In order to avoid damage to the converter of wind generators due to voltage rise, the grid codes for wind power integration require wind generators to carry out the high voltage

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ride through. Wind generators are even cut off when the access point voltage is higher. CF produces a cascading reaction in the inverter station, and may lead to the high voltage ride through of wind generators at the sending-end grid. The overvoltage caused by CF leads to wind generators temporarily invalid or tripping, resulting in an inter-regional cascading reaction [5,6]. The interruption of the CF caused cascading reaction is a new problem faced by the wind power transmission system via LCC-HVDC.

Many researchers have focused on the CF prevention of LCC-HVDC. Some researchers are committed to improving CF prevention control (CFPREV) [7–9]. However, it is difficult to accurately determine the starting voltage and firing angle adjustment of CFPREV since CF is affected by many factors. Therefore, a predictive method of SCF is proposed based on the derivation of critical commutation voltage of SCF [26]. However, the derived critical commutation voltage can only be used to judge whether SCF occurs and provide a basis for suppressing SCF, but cannot reflect the degree of overvoltage in the rectifier station. Researchers have proposed to suppress SCF by adding additional reactive power compensation devices [10], energy storage [11], transformation of converter topologies [12] and optimization of control system [13–15]. However, the first three methods are still difficult to be applied in industry due to the increased complexity of control system and investment cost. Some researchers proposed to suppress SCF by using additional virtual resistors [13] and boosting the starting voltage of voltage-dependent current order limiter (VDCOL) [14]. In Wang et al. [15], a SCF suppression method based on real-time reactive power demand to determine the DC current command value is proposed. However, the reactive power consumption of both the rectifier station and the inverter station of LCC-HVDC is affected by the DC current. The existing methods only focus on the recovery of CF in the inverter station, lack of consideration of overvoltage of the rectifier station. It is not conducive to the overvoltage suppression in the rectifier station, and may even aggravate the overvoltage and exacerbate the cascading reaction of CF at the sending-end grid.

The research on avoiding the high voltage ride through of wind generators at the sending-end grid caused by CF focuses on improving the ride through strategy of wind generators [16–18] and installing reactive power compensation device [19]. However, the ride through strategy is difficult to solve the overvoltage from the basics. The reactive power compensation device can suppress the overvoltage, but there are problems such as high cost and difficulty in coordination. Adjusting the DC current of LCC-HVDC to balance the reactive power of rectifier station is an immediate measure to avoid the high voltage ride through of wind generators. An improved method of VDCOL is proposed to suppress overvoltage of rectifier station in Wang et al. [20]. A constant reactive power control to inhibit overvoltage is proposed based on the overvoltage calculation in Yin and Li [21], but it only takes overvoltage suppression as the control objective. The influence of reactive power control of rectifier station on CF and transmitted active power is not considered. However, the DC current, DC voltage and firing angle of rectifier station and inverter station affect each other. The regulation of reactive power or DC current affects the overvoltage of rectifier station, but also changes the CF recovery characteristics of inverter station and the active power transmitted by LCC-HVDC. Therefore, the existing methods ignore the impact on the inverter station, which affects the transmitted active power, and may exacerbate the CF caused cascading reaction in the inverter station.

The control with the independent goal of CF suppression in the inverter station or overvoltage control in the rectifier station is difficult to take into account the CF caused cascading reaction of the inverter station and that of sending-end grid, and may even aggravate the cascading process. Therefore, this paper proposes a new method to interrupt the CF caused cascading reaction at the sending and receiving ends through DC current control of LCC-HVDC. The coordination of the exchanged reactive power of the rectifier station and the transmitted power can be realized on the premise of avoiding the CF caused cascading reaction through adaptive DC current control. Then, the overvoltage of the rectifier station is suppressed, and the CF caused cascading reaction at the sending-end grid can be avoided. The key technical innovations of this work are reflected in the following aspects:

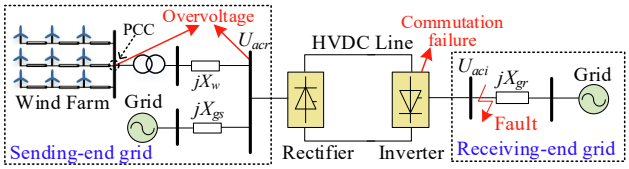


Fig. 1. Structure diagram of HVDC system with wind farm integration.

1) The cascading reaction process in wind power transmission system via HVDC caused by commutation failure is elaborated.
2) The feasible range of DC current to avoid the CF caused cascading reaction is quantified by analyzing the relationship between the DC current, AC voltage of the rectifier station and inverter extinction angle.
3) The feasible power region of the inverter station which can suppress the CF caused cascading reaction is constructed.
4) An adaptive DC current control to interrupt the CF caused cascading reaction of the inverter station and that of sending-end grid simultaneously is proposed based on the feasible power region.

This paper is organized as follows. Section II analyzes the CF caused cascading reaction in the wind power transmission system via LCC-HVDC. In Section III, the analytical expression of DC current for avoiding the high voltage ride through of wind generators at the sending-end grid and the CF of the inverter station is deduced; then the feasible power region to suppress the CF caused cascading reaction is established. Section IV proposes the adaptive current control principle based on the feasible power region. Section V proposes the command value calculation and implementation method of adaptive DC current control. Section VI uses CIGRE standard test system to verify the correctness of the theoretical analysis.

2. Commutation failure caused cascading reaction

The HVDC system with wind farm integration can be equivalent to Fig. 1. The LCC-HVDC converter station is composed of converter transformer, converter, filter, control and protection device, etc. The rectifier station and inverter station are collectively called the converter station. The inverter station adopts a semi-controlled thyristor, which only relies on the grid voltage to restore the blocking ability. The internal stored charge generated during the forward conduction interval must be removed before the valve can establish the forward voltage blocking ability, so as to ensure the success of commutation. When a converter valve that is supposed to turn off, continues to conduct without transferring its current to the next valve in the firing sequence, the commutation failure occurs. The amplitude drop and waveform distortion of commutation voltage caused by AC fault are easy to cause CF. In order to ensure the normal operation after CF, the inverter station is generally equipped with constant extinction angle control (CEAC), constant current control (CCC), current error control (CEC) and VDCOL. And in the CF process, the rectifier station adopts constant current control (CCC) [20]. The control system adjusts the DC current, DC voltage, extinction angle and other electrical quantities to protect the safety of the converter valve and promote the recovery of the inverter station.

In the process of CF and its recovery, the reactive power consumed by the converter station fluctuates greatly with the changes of DC current, DC voltage and firing angle, which leads to the fluctuation of AC bus voltage of the converter station. The active power and reactive power consumed by the converter station can be expressed as:

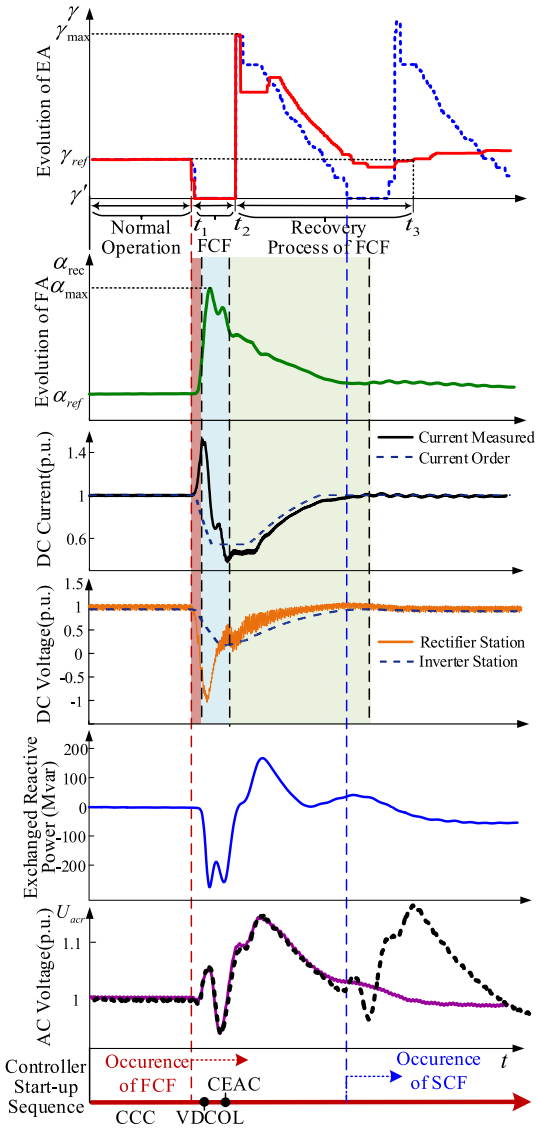


Fig. 2. CF and its recovery process.

$$\begin{cases} P_d = U_d I_d \\ Q_d = I_d \sqrt{U_{d0}^2 - U_d^2} \end{cases} \tag{1}$$

where, I_d is the DC current; U_d is the DC voltage of converter station; U_{d0} is the ideal no-load DC voltage and is expressed as $U_{d0} = 3\sqrt{2}kNU_{ac}/\pi$.

The dynamic changes of the electrical quantities of converter stations during CF is tested based on CIGRE-HVDC benchmark mode. The wind farm is connected to the sending-end grid with a capacity of 500 MW. The dynamic changes of the extinction angle (EA) and DC voltage of inverter station, firing angle, DC current, DC voltage, exchanged reactive power and AC bus voltage of rectifier station after CF with the effect of the control system is shown in Fig. 2. The process can be divided into three stages. Stage 1 is the DC current rising stage, as represented in the red area in Fig. 2. The FCF caused by the AC fault at the receiving-end grid occurs at the inverter station at t_1. During CF, the DC side of the converter valve of the inverter station is short-circuited, resulting in a rapid increase in DC current and a drop in DC voltage. In the interim, the reactive power consumed by the rectifier station increases significantly, and the AC bus voltage of the rectifier station decreases.

Stage 2 is the DC current drop stage, as shown in blue area in Fig. 2. The firing angle of rectifier station increases rapidly to reduce the DC current under the CCC of rectifier station. When the DC voltage of the inverter station is lower than the starting voltage of VDCOL, VDCOL starts and reduces the DC current according to a preset characteristic curve. The decrease of DC current leads to the increase of CCC input of the rectifier station, which further increases the firing angle of the rectifier station. As the DC current decreases, the difference between the DC current command value and the DC current decreases, the firing angle of the CCC of the rectifier station gradually decreases, and the DC voltage of the rectifier station gradually increases. The reactive power consumed by the rectifier station decreases with the rise of DC voltage and the decrease of DC current according to (1). As the switching time constant of the reactive power compensation device in the rectifier station is at least 0.1 s, and greater than the response time of the control system for tens of milliseconds [21]. Thus, the surplus reactive power is produced in the rectifier station, resulting in overvoltage on the AC bus of the rectifier station.

Stage 3 is the DC current recovery stage, as shown in the green area in Fig. 2. The DC voltages of rectifier station and inverter station rise gradually by the effect of the control system. The DC current increases linearly with the recovery of DC voltage under the VDCOL. The firing angle of rectifier station decreases gradually with the recovery of DC current under the CCC of rectifier station. The EA firstly transitions and then gradually recovers to the steady state value under the alternating effect of CEAC and CCC of inverter station.

The reactive power consumed by the rectifier station fluctuates greatly after the CF of the inverter station caused by the AC fault of receiving-end grid, and the AC bus voltage of the rectifier station decreases first and then increases. In the rising stage of AC bus voltage of the rectifier station, the access point voltage of wind farms in the near area rises due to the conduction of power and voltage. According to the high voltage ride through criteria of wind generators [22], when the access point voltage of wind generators is greater than the threshold value of high voltage ride through and exceeds the ability of continuous operation without off-grid, the tripping of wind generators occurs. The reactive power compensation device near the wind farm cannot be cut off in time after the tripping of some wind generators, and the surplus reactive power will further raise the overvoltage and expand the tripping scope. If the AC fault causing CF is not cleared in time or the control effect of the LCC-HVDC is improper, SCF is prone to occur during the recovery of FCF, as shown in the EA of the blue dotted line in Fig. 2. SCF leads to another sudden change in power, causing multiple overvoltage in the AC bus of the rectifier station, as shown by the AC voltage of the black dotted line in Fig. 2. The tripping of wind generators at the sending-end due to overvoltage may occur again. A complex cascading reaction process is triggered by the CF of the inverter station caused by the AC fault of the receiving-end grid.

3. Modelling of feasible power region

3.1. Feasible range of DC current

During CF and its recovery, the reactive power consumed by the rectifier station is determined by the DC current, that is, the DC current is closely related to the AC bus voltage of the rectifier station. According to the commutation process of the converter, the DC voltage of the rectifier station can be expressed as [20]:

$$U_{dr} = N_1 \left(\frac{3\sqrt{2}\, k_1 U_{acr}}{\pi} \cos\alpha_{\mathrm{rec}} - \frac{3}{\pi} X_{\mathrm{r1}} I_d \right) \tag{2}$$

where, α_{rec} is the firing angle of rectifier station; U_{acr} is the RMS of AC bus voltage of the rectifier station; k_1 is the ratio of the converter transformer in the rectifier station; N_1 is the number of 6-pulse converters per pole in the rectifier station; X_{r1} is the equivalent converter reactance of the rectifier station.

According to Eq. (2) and the voltage drop equation of DC circuit [21],

the AC bus voltage of the rectifier station can be written as:

$$U_{acr} = \frac{(U_{di} + I_d R_d)\pi + 3N_1 X_{r1} I_d}{3\sqrt{2}k_1 N_1 \cos\alpha_{rec}} \tag{3}$$

where, U_{di} is the DC voltage of inverter station; R_d is the equivalent resistance of DC transmission lines.

The surplus reactive power at the AC bus of the rectifier station is

$$\Delta Q = Q_{cr} + Q_{AC} - Q_{dr} \tag{4}$$

where, Q_{dr} is the reactive power consumed by rectifier station; Q_{cr} is the compensated reactive power of rectifier station; Q_{AC} is the exchanged reactive power between the rectifier station and the sending-end grid.

According to (1), the reactive power consumed by the rectifier station is proportional to the DC current. Therefore, the decrease of DC current increases the surplus reactive power, resulting in the increase of the rise amplitude of the AC bus voltage in the rectifier station. When the AC bus voltage of the rectifier station is less than the threshold value of high voltage ride through, the high voltage ride through of wind generators can be avoided. Therefore, according to Eqs. (1) and (3), the lower limit of DC current to avoid the of high voltage ride through of wind generators at the sending-end grid can be obtained:

$$I_{d,v} = \frac{\frac{3\sqrt{2}k_1 N_1 U_{acrN} U_{pcc,\max}}{\pi}\cos\alpha_{rec} - U_{di}}{R_d + \frac{3}{\pi}N_1 X_{r1}} \tag{5}$$

where, $U_{pcc,\max}$ is the threshold value of high voltage ride through of wind generators in per unit. α_{rec} is the firing angle of rectifier station and can be expressed as:

$$\alpha_{rec} = \pi - K_p\left[(I_{d\text{ - ord}} - I_{d\text{ - rec}}) + \frac{1}{T_i}\int (I_{d\text{ - ord}} - I_{d\text{ - rec}})dt\right] \tag{6}$$

where, K_p and T_i are the proportional coefficient and integral time constant of the CCC of the rectifier station;$I_{d\text{ - ord}}$ and $I_{d\text{ - rec}}$ are the command value and measured value of DC current of rectifier station, respectively.

In terms of the commutation voltage–time area [23], the DC current and commutation voltage of the inverter station during commutation process meet the following relationship:

$$\int_{\pi-\beta}^{\pi-\gamma}\left(\sqrt{2}k_2 U_{aci}\sin\omega t\right)d\omega t = X_{r2}[I_d(\pi-\gamma) + I_d(\pi-\beta)] \tag{7}$$

where, U_{aci} is the RMS of AC bus voltage of inverter station; k_2 is the ratio of the converter transformer of inverter station; X_{r2} is the equivalent converter reactance of the inverter station; γ is the EA of inverter station; β is the leading firing angle of inverter station.

In order to avoid SCF of inverter station, the EA should always be greater than the critical EA during the CF recovery. Therefore, the upper limit of DC current to avoid SCF can be obtained according to Eq. (7):

$$I_{d,c} = \frac{\sqrt{2}k_2 U_{aci}(cos(\gamma_{\lim} + \phi) - \cos(\gamma_{\lim} + \mu))}{2X_{r2}} \tag{8}$$

Where, $\gamma_{\lim}$ is the critical EA of CF, corresponding to the deionization recovery time of thyristor. ϕ is the line-to-line commutating voltage zero crossing phase shift.

The feasible range of DC current to avoid the high voltage ride through of wind generators at the sending-end grid and SCF in the inverter station is:

$$I_{d,v} < I_d < I_{d,c} \tag{9}$$

The DC current satisfying Eq. (9) can avoid the cascading reaction at the inverter station and the sending-end grid caused by the CF of inverter station under the AC fault of the receiving-end grid. The lower limit of DC current increases with the decrease of DC voltage of inverter station. The upper limit of DC current decreases with the decrease of AC bus voltage of inverter station. Therefore, the feasible range of DC current is affected by the severity of the AC fault.

3.2. Feasible power region of inverter station

According to (1) and (2), the active power and reactive power consumed by the inverter station are related to the electrical quantities, DC current, EA, firing angle and AC bus voltage. By combining (1) and the DC voltage equation of the inverter station, the output active power and exchanged reactive power of the inverter station can be obtained as follows:

$$P_d(U_{aci}, I_d) = N_2\left(\frac{3\sqrt{2}\,U_{aci}}{k_2\pi}\cos\gamma - \frac{3}{\pi}X_{r2}I_d\right)I_d \tag{10}$$

$$Q_{aci}(U_{aci}, I_d) = Q_{ci} - \left[\left(\frac{3\sqrt{2}N_2}{k_2\pi}\right)^2(1-\cos^2\gamma)U_{aci}^2 I_d^2 - \left(\frac{3}{\pi}N_2 X_{r2}\right)^2 I_d^4 + \frac{18\sqrt{2}X_{r2}N_2^2\cos\gamma}{k_2\pi^2}U_{aci}I_d^3\right]^{1/2} \tag{11}$$

where, N_2 is the number of 6-pulse converters per pole in the inverter station; Q_{ci} is the reactive power provided by reactive power compensation device of inverter station. The transmission from the inverter station to the AC grid is set as the positive direction.

The derivative of exchanged reactive power of (11) against DC current can be obtained:

$$\frac{\partial Q_{aci}}{\partial I_d} = D_I\left[\left(\frac{6N_2}{k_2\pi}\right)^2 U_{aci}^2 I_d - \left(\frac{6N_2}{k_2\pi}\cos\gamma\right)^2 U_{aci}^2 I_d - \left(\frac{6}{\pi}N_2 X_r\right)^2 I_d^3 + \frac{54\sqrt{2}X_{r2}N_2^2\cos\gamma}{k_2\pi^2}U_{aci}I_d^2\right] \tag{12}$$

where,

$$D_I = -\frac{1}{2}\left[\left(\frac{3\sqrt{2}N_2}{k_2\pi}\right)^2 U_{aci}^2 I_d^2 - \left(\frac{3\sqrt{2}N_2}{k_2\pi}\cos\gamma\right)^2 U_{aci}^2 I_d^2 - \left(\frac{3}{\pi}N_2 X_{r2}\right)^2 I_d^4 + \frac{18\sqrt{2}X_{r2}N_2^2\cos\gamma}{k_2\pi^2}U_{aci}I_d^3\right]^{-1/2} \tag{13}$$

The derivative of exchanged reactive power of (11) against EA can be obtained:

$$\frac{\partial Q_{aci}}{\partial\gamma} = -\frac{9D_r N_2^2 U_{aci} I_d^2 \sin\gamma}{k_2\pi^2}\left(\frac{U_{aci}}{k_2}\cos\gamma - \frac{\sqrt{2}}{2}X_{r2}I_d\right) \tag{14}$$

where,

$$D_r = \left[\left(\frac{3\sqrt{2}N_2}{k_1\pi}\right)^2 U_{aci}^2 I_d^2 - \left(\frac{3\sqrt{2}N_2}{k_2\pi}\cos\gamma\right)^2 U_{aci}^2 I_d^2 + \frac{18\sqrt{2}X_{r2}N_2^2\cos\gamma}{k_2\pi^2}U_{aci}I_d^3 - \left(\frac{3}{\pi}N_2 X_{r2}\right)^2 I_d^4\right]^{-1/2} \tag{15}$$

The calculated coefficients meet $D_I < 0, D_r > 0$ according to the value range of AC bus voltage, DC current, EA, ratio of the converter transformer and converter reactance. Therefore, Eq.(12) and (14) are both less than 0. The exchanged reactive power increases with the decrease of DC current and EA. The active power increases with the increase of DC current and decreases with the increase of EA according to (10).

In order to ensure the normal commutation of the converter, the EA meets:

$$\gamma_{\lim} \leqslant \gamma \leqslant \gamma_{ref} \tag{16}$$

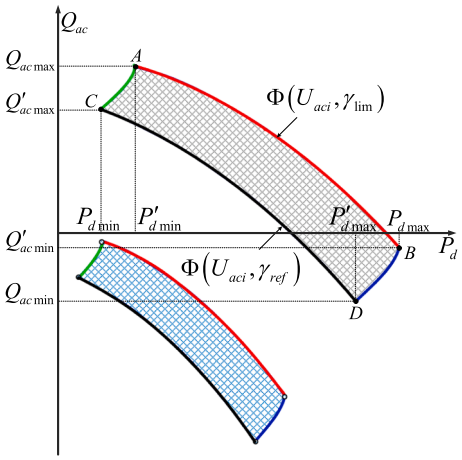


Fig. 3. Feasible power region of inverter station.

where,γ_{ref} is the initial reference value of extinction angle of inverter station.

By combining (10) and (11), the relationship function between the active power transmitted by inverter station and exchanged reactive power under a certain AC bus voltage can be written:

$$Q_{aci} - B_c U_{ac}^2 + F\left(p_1 U_{aci}\cos\gamma - p_2\sqrt{(p_3 U_{aci}\cos\gamma)^2 - p_4 P_d}\right) = 0 \tag{17}$$

where, the coefficient meets the following expression:

$$F^2 = p_3^2 U_{aci}^2 - 0.5p_3^2 U_{aci}^2\cos^2\gamma + 0.25p_4 dP_d$$
$$- 0.5p_3 U_{aci}\cos\gamma\sqrt{(p_3 U_{aci}\cos\gamma)^2 - p_4 P_d}$$

where, the coefficients are given by:

$$p_1 = \frac{\sqrt{2}}{2k_2 X_{r2}}, p_2 = \frac{\pi}{6N_2 X_{r2}}, p_3 = \frac{3\sqrt{2}N_2}{k_2\pi}, p_4 = \frac{12N_2 X_{r2}}{\pi}$$

The feasible range of active power and exchanged reactive power can be obtained by substituting (9) into (10) and (11), and combining (16) and (17). The feasible range can simultaneously meet the normal commutation of inverter and avoid the high voltage ride through of wind generators at the sending-end grid, that is, the feasible power region of inverter station that can suppress CF caused cascading reaction.

The green curve *AC* in Fig. 3 is the boundary of the feasible power region that meets the lower limit of the DC current during normal commutation of inverter, which can be obtained by substituting (5) and (16) into (10) and (11). The blue curve *BD* in Fig. 3 is the boundary of the feasible power region corresponding to the upper limit of the DC current during normal commutation of inverter, which can be obtained by substituting (8) and (16) into (10) and (11). The power curve $\gamma_{\lim}$is determined by (9), (10) and (17). The red curve *AB* and the black curve *CD* are the power curves under $\gamma_{\lim}$ andγ_{ref}, respectively, forming the upper and lower boundaries of the feasible power region. The shaded area enclosed by the four curves is the feasible power region of the inverter station. In Fig. 3, the reactive power at point *A*,$Q_{ac\max}$, is the maximum exchanged reactive power corresponding to the lower limit of DC current and the critical value of EA. The reactive power at point *D*,$Q_{ac\min}$, is the minimum exchanged reactive power corresponding to the upper limit of DC current and the initial reference value of EA.

The output active power and exchanged reactive power of the inverter station decrease with the reduction of the AC bus voltage of the inverter station according to (10) and (11). The feasible power region moves from the first quadrant to the fourth quadrant of the active-reactive coordinate system, as shown in the gray and blue shaded areas in Fig. 3. The upper limit and lower limit of DC current change with the AC bus voltage of the inverter station. The range and distribution of the feasible power region depend on the AC bus voltage of the inverter station. The minimum and maximum exchanged reactive power within the feasible power region change with the upper and lower limits of DC current. The maximum exchanged reactive power may be positive or negative under different AC bus voltage of the inverter station.

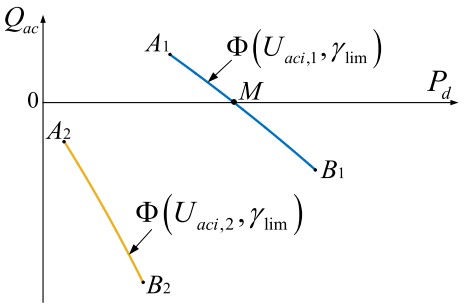


Fig. 4. DC current control principle.

4. Control principle for CF caused cascading reaction

The feasible power region depends on the AC bus voltage and DC voltage of the inverter station. The feasible power region can be determined after the AC fault at the receiving-end grid. And the DC current can be controlled according to the feasible power region, so as to avoid the SCF and high voltage ride through of wind generators at the sending-end grid. It is unfavorable to the voltage stability of the AC grid and the recovery of the inverter station when the inverter station absorbs a large amount reactive power from the grid after AC fault. This situation should be avoided, that is, to avoid the exchanged reactive power being less than zero [13]. The exchanged reactive power of the inverter station increases with the reduction of DC current according to Eqs. (10) and (12). However, the active power transmission is limited when reducing the DC current, and may aggravate the power shortage of the grid and threaten the stability of the grid frequency and power angle. The active power and exchanged reactive power of the inverter station should be coordinated within the feasible power region of the inverter station under the condition of avoiding CF caused cascading reaction. Therefore, the primary objective of DC current control is to make the exchanged reactive power greater than zero, and the secondary objective is to maximize the active power.

The feasible power region moves from the first quadrant to the fourth quadrant with the reduction of the AC bus voltage of the inverter station. The blue and orange curves in Fig. 4 are the upper boundary of the feasible power region under different AC bus voltages of inverter station, where$U_{aci,1} > U_{aci,2}$. When the maximum exchanged reactive power in the feasible power region is greater than 0 and the minimum exchanged reactive power is less than 0, the upper boundary of the feasible power region is located in the first and fourth quadrants, as represented by the blue line in Fig. 4. The point *M* is the point with zero exchanged reactive power under$U_{aci,1}$. And the point *M* is the power operation point of inverter station with exchanged reactive power not less than 0 and maximum active power under the condition of avoiding CF caused cascading reaction. The DC current shall be controlled to make the inverter station operate at point *M*.

When the maximum exchanged reactive power within the feasible power region of the inverter station is less than 0, the upper boundary of the feasible power region is located in the fourth quadrant, as shown by the orange curve in Fig. 4. Since the exchanged reactive power that can avoid the CF caused cascading reaction is always less than zero, the DC current should be controlled to minimum absorbed reactive power from the receiving-end grid. The DC current shall be controlled to make the inverter station operate at point A_2. The current at point A_2 is the lower

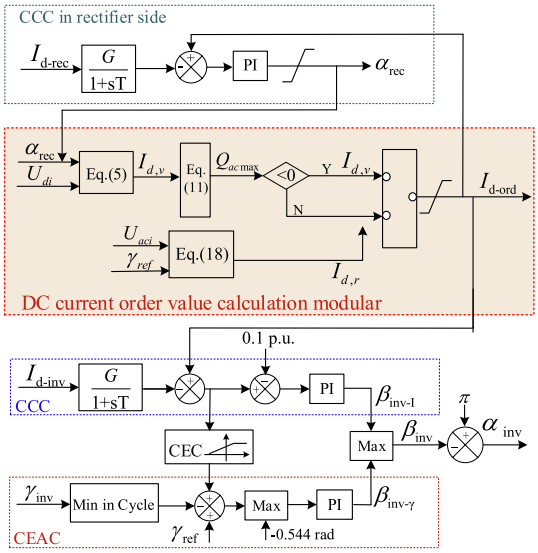


Fig. 5. Block diagram of adaptive current control.

limit of DC current to avoid the high voltage ride through of wind generators at the sending-end grid. The migration of power operation point is achieved through the adaptive DC current control according to the AC bus voltage of inverter station. The adaptive DC current control can restrain the CF caused cascading reaction and consider the stability and recovery of the power grid under fault.

5. Adaptive current control based on feasible power region

5.1. Command value

The DC current command value can be calculated by (5) when the feasible power region is in the fourth quadrant. When the feasible power region of the inverter station is located in the first and fourth quadrants, the DC current command value can be obtained by making the exchanged reactive power of Eq. (11) equal to zero:

$$I_{d,r}=\frac{-n+\sqrt{n^2-\frac{8}{3}mp+4m(A+B)}-\sqrt{C+D}}{4m} \tag{18}$$

where

$$A=\sqrt[3]{\left[\left(\frac{2p^3+27n^2q-72mpq}{54}\right)^2-\left(\frac{p^2+12mq}{9}\right)^3\right]^{1/2}+\frac{2p^3+27n^2q-72mpq}{54}}$$

$$B=\frac{P^2+12mq}{9A},\ C=2n^2-\frac{16}{3}mp-4m(A+B),$$

$$D=\frac{8mnp-2n^3}{\sqrt{n^2-\frac{8}{3}mp+4m(A+B)}}$$

where, the coefficients m, n, p and q are given by

$$m=\frac{9N_2^2X_{r2}^2}{\pi^2},n=-\frac{18\sqrt{2}k_2X_{r2}N_2^2\cos\gamma_{ref}}{\pi^2}U_{aci},$$

$$p=\frac{18k_2^2N_2^2\left(\cos^2\gamma_{ref}-1\right)}{\pi^2}U_{aci}^2,\ q=Q_{ci}^2$$

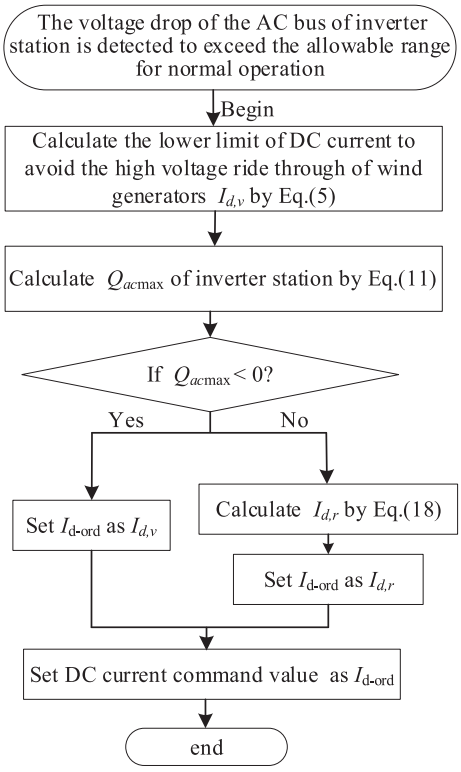


Fig. 6. Flowchart of proposed method.

Table 1
Gain and time constant values of controllers.

Controller	Measurement Link		PI controllers	
	Gain	Time constant/s	Proportional Gain	Integral Time Constant/s
CCC in rectifier side	0.5	0.0012	1.0986	0.01092
CCC in inverter side	0.5	0.0012	0.6300	0.01524
CEAC in inverter side	_	_	0.7506	0.0544

5.2. Control implementation

The adaptive current control method based on the feasible power region is displayed in Fig. 5. And the flow chart of control method is illustrated in Fig. 6. When the voltage drop of the AC bus of the inverter station is detected to exceed the allowable range for normal operation, the adaptive current control starts. The lower limit of DC current $I_{d,v}$ to avoid the high voltage ride through of wind generators at the sending-end grid can be calculated by substituting the DC voltage of the inverter station and the firing angle of the rectifier station after the fault into (5). In order to avoid the influence of transmission delay, the firing angle output of the CCC of the rectifier station in the previous sampling period is adopted when calculating the lower limit of DC current in each sampling period. Then, the maximum exchanged reactive power of the inverter station is calculated by taking the AC bus voltage U_{aci} and $I_{d,v}$ into (11). When the maximum exchanged reactive power is greater than 0, the DC current command value is determined by taking the AC bus voltage of inverter station U_{aci} and the initial reference value of EA γ_{ref} into Eq. (18). When the maximum exchanged reactive power is less than 0, the DC current command value is set to$I_{d,v}$.

The DC currents of the rectifier station and inverter station are measured. The time constant and the gain values of measurement links

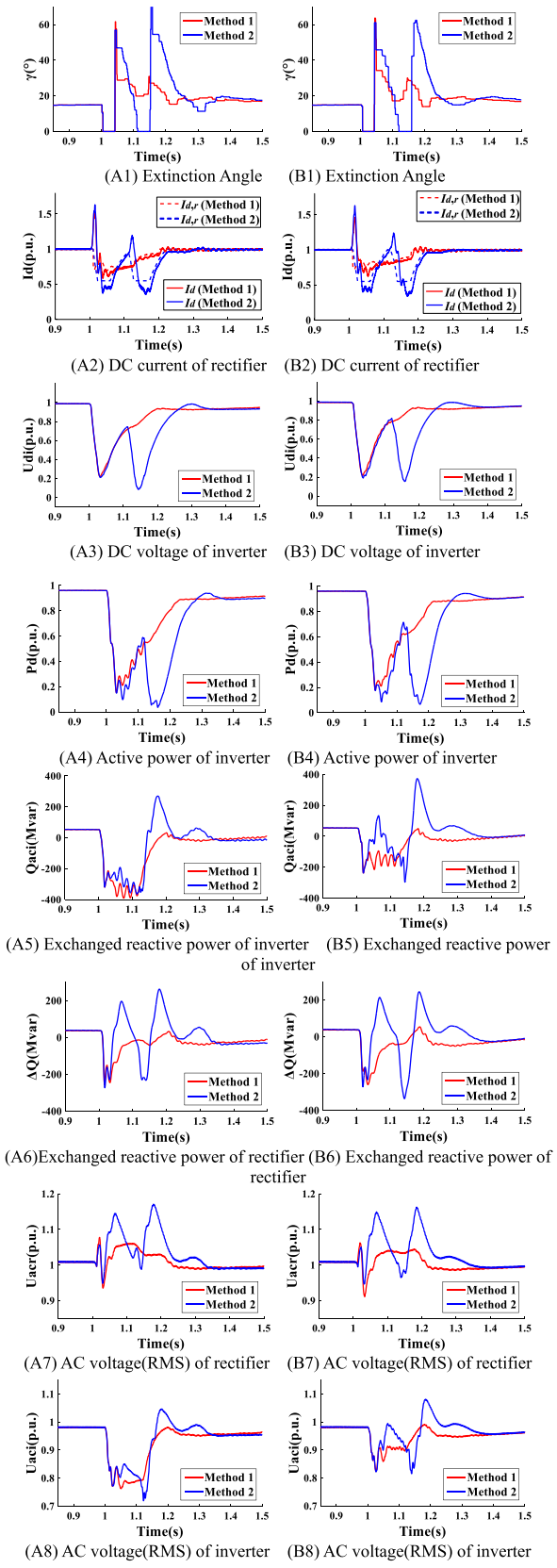


Fig. 7. Electrical quantities of LCC-HVDC under different fault types: (A1-A8), TPG fault and $t_f = 0.1$ s; (B1-B8), SPG fault and $t_f = 0.1$ s.

are enumerated in Table 1. And the proportional gain and time constant values of each PI controller in Fig. 5 are tabulated in Table 1. The difference between the command value and measurement value of DC current of the rectifier station is used as the input of the CCC of the rectifier station. The PI controller outputs the firing angle to realize the regulation of the DC current at the rectifier side. The difference between the command value and measurement value of DC current of the inverter station is taken as the input of the PI controller in the CCC of the inverter station. The PI controller outputs the leading firing angle, and then it is compared with the output of CEAC to obtain the firing angle command value of the inverter station, so as to realize the adjustment of DC current and EA of the inverter station. When the deviation between the DC current of the inverter station and the DC current command value of the rectifier station is 0.1p.u., the CEC is activated to realize the smooth switching between the CCC and CEAC at the inverter station. The function of the maximum value link in the CEAC is to limit the maximum value of EA [20]. The relevant calculation and instruction judgment module of the controller starts after the fault, and can be realized based on the digital signal processor with the processing speed of microsecond [9,24]. The data acquisition time is about 10 ms. Considering that the time interval from fault to SCF and overvoltage at the sending-end gird is tens of milliseconds [25], the adaptive DC current control has sufficient response time to interrupt the CF caused cascading reaction.

6. Case studies

PSCAD/EMTDC software is used to verify the correctness of proposed theoretical analysis in this paper. LCC-HVDC adopts CIGRE Benchmark HVDC model as test system [26]. The mono-polar HVDC is equipped with 12-pulse converters on both rectifier and inverter stations with rated voltage 500 kV. The rated capacity is 1000 MW and the converter reactance 13.5 Ω. The inverter and rectifier capaity is 1000 MW. The Short-Circuit Ratio (SCR) of AC grid at rectifier station and inverter station are $2.5\angle 84^\circ$ and $2.5\angle 75^\circ$, respectively. Under normal operation, the DC current is 2 kA, and EA is 15.2° at the inverter side; the rated voltage of AC bus of rectifier station is 345 kV, and the rated voltage of AC bus of inverter station is 230 kV. The rated capacity of the wind farm is 500 MW. The threshold value of high voltage ride through of wind generators is 1.1p.u. [22]. The effects of the following two control methods are compared by setting different fault types and fault degrees: Method 1 is the adaptive current control method proposed in this paper. Method 2 is the VDCOL in CIGRE Benchmark HVDC model.

Different types of faults (including three-phase(TP) fault, single phase(SP) fault) are set at the AC bus of the inverter station. The transition inductance is 0.30H, and the fault occurrence time is 1 s with the fault duration of 100 ms. The AC bus voltages of the inverter station drop to 0.774p.u and 0.823p.u. after TP and SP faults, respectively. The maximum exchanged reactive power of the inverter station within the feasible power region is greater than 0 according to (11). Therefore, the DC current command value of Method 1 is calculated by (18).

Fig. 7 exhibits the electrical quantities of LCC-HVDC under different fault types. The electrical quantities when using Method 1 and Method 2 are shown as the red and blue lines in Fig. 7 (A1-A8) and (B1-B8), respectively.

EA drops to 0° after the fault, and the FCF occurs as represented in Fig. 7 (A1, B1). The DC current command value of Method 2 is slightly higher than that of Method 1 after the fault is cleared, resulting in the increase of reactive power absorbed by the inverter station from the AC grid, as shown in Fig. 7 (A2, A5) and (B2, B5). EA drops to 0° again about 100 ms after the FCF under Method 2, and SCF occurs. And the DC voltage and DC current present two large changes as shown in Fig. 7 (A2, A3) and (B2, B3). The generated reactive power shock then leads to large fluctuations in the AC bus voltage of the inverter station, as shown in Fig. 7 (A8, B8). However, no SCF occurs in the inverter station under Method 1, and the reactive power shock and voltage fluctuation are

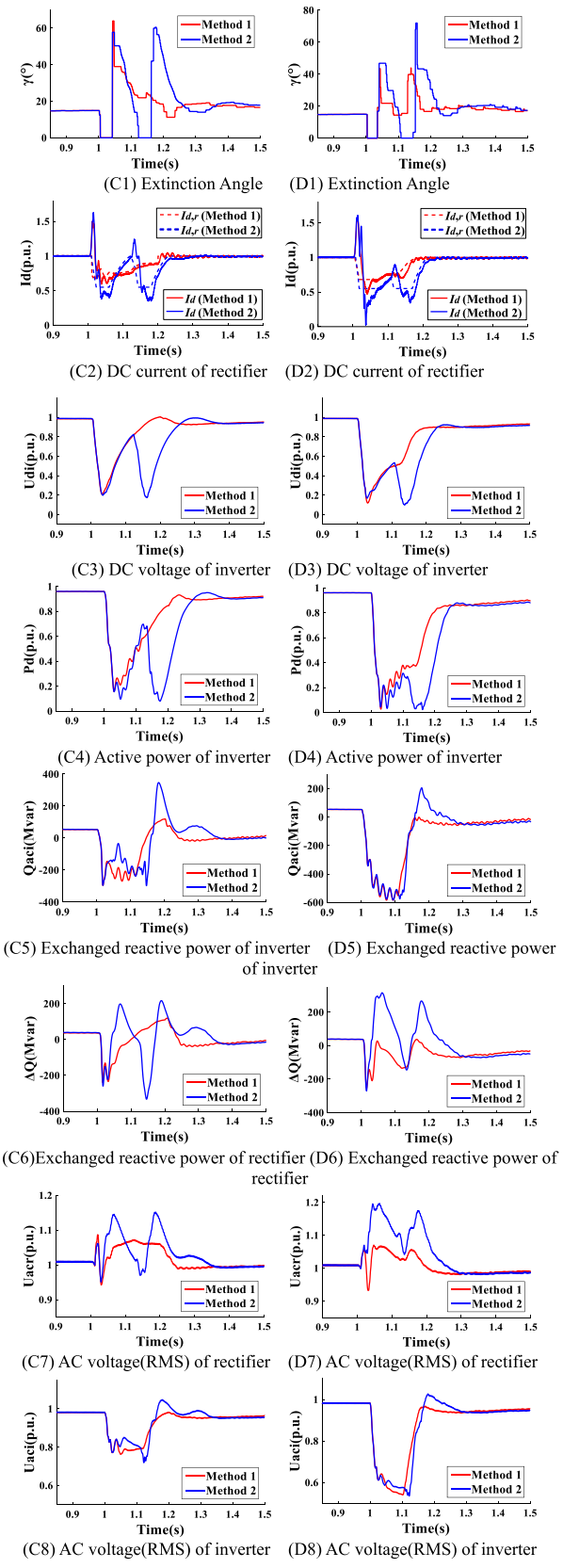


Fig. 8. Electrical quantities of LCC-HVDC under different fault depth: (C1-C8), $L_f = 0.5$H.and $t_f = 0.1$ s; (D1-D8), $L_f = 0.1$H.and $t_f = 0.1$ s.

effectively alleviated. Meanwhile, the DC current command value and the measured value under Method 1 during the FCF recovery are more stable than those under Method 2, and the valley value is larger than that of Method 2 as shown in Fig. 7 (A2, B2). Under Method 1, the active power of the inverter station during the CF recovery process is significantly higher than that of control method 2, as shown in Fig. 7 (A4, B4). The transmission of active power is guaranteed to the greatest extent while suppressing SCF, and the recovery of AC bus voltage of inverter station is accelerated under Method 1.

The DC current command value and measured value of the rectifier station under Method 1 are greater than those under Method 2 during CF recovery, as shown in Fig. 7, and the exchanged reactive power of the rectifier station is significantly lower than that of Method 2, as shown in Fig. 7 (A6, B6). Under Method 2, the most serious overvoltage peak of the AC bus of the rectifier station caused by the TP fault at the receiving-end grid is 1.146p.u. at 1.067 s, while the most serious overvoltage peak caused by the SP fault is 1.160p.u., as shown in Fig. 7 (A7, B7). The voltages are higher than the threshold value of high voltage ride through of wind generators, causing the control switching of wind generators and entering the high voltage ride through. The AC bus voltage of the rectifier station exceeds 1.1p.u. again after the occurrence of SCF. In contrast, it can be seen from Fig. 7 (A7, B7) that under Method 1, the peak values of the AC bus voltage of the rectifier station under TP fault and SP fault are 1.058p.u. and 1.036p.u., respectively. The overvoltage of the rectifier station can be suppressed under the proposed method, which avoids the high voltage ride through of wind generators at the sending-end grid.

The fault transition inductance is changed to 0.5H and 0.1H. The three-phase fault occurs at the AC bus of the inverter station at 1 s, with fault duration of 100 ms. When the transition inductance is 0.5H, the AC bus voltage of the inverter station drops to 0.806p.u. after the fault. The maximum exchanged reactive power of the inverter station in the feasible power region is greater than 0 according to (11). Thus, the DC current command value of Method 1 is calculated by (18). When the transition inductance is 0.1H, the AC bus voltage of the inverter station drops to 0.625p.u. after the fault. And the maximum exchanged reactive power of the inverter station in the feasible power region is less than 0 according to (11). Thus, the DC current command value of Method 1 is calculated by (5).

Fig. 8 exhibits the electrical quantities of LCC-HVDC under different fault depth. The electrical quantities when using Method 1 and Method 2 are shown as the red and blue lines in Fig. 8 (C1-C8) and (D1-D8), respectively.

EA drops to 0° after the fault, and the FCF occurs as represented in Fig. 8 (C1, D1). The DC current command value changes with the DC voltage under Method 2, and two times of DC current large changes can be observed as shown in Fig. 8 (C2, C3) and (D2, D3). This leads to two large fluctuations in the exchanged reactive power, active power and AC bus voltage of the inverter station. EA drops to 0° again about 100 ms after the FCF under Method 2, and SCF occurs. However, no SCF occurs in the inverter station under Method 1. During the CF recovery, the DC current command value and measured value of Method 1 are slightly higher than that of Method 2. The simulation results in Fig. 8 (C4, D4) indicate that the active power transmitted by the inverter station under Method 1 is greater than that of Method 2. And the recovery speed of active power transmission is effectively improved.

Meanwhile, the exchanged reactive power of rectifier station under Method 1 is significantly lower than that of Method 2 during CF recovery, as shown in Fig. 8 (C6, D6). The peak value of AC bus voltage of rectifier station is reduced from 1.149p.u. with $L_f = 0.5$H and 1.197p.u. with $L_f = 0.1$H to 1.042p.u. and 1.060p.u., respectively. The SCF is effectively prevented and the overvoltage at the AC bus of the rectifier station is suppressed under Method 1, and the recovery of the AC bus voltage at the inverter station and DC transmission power is promoted.

In order to further validate the performance of the proposed control method under real-time conditions, a hardware simulation platform for

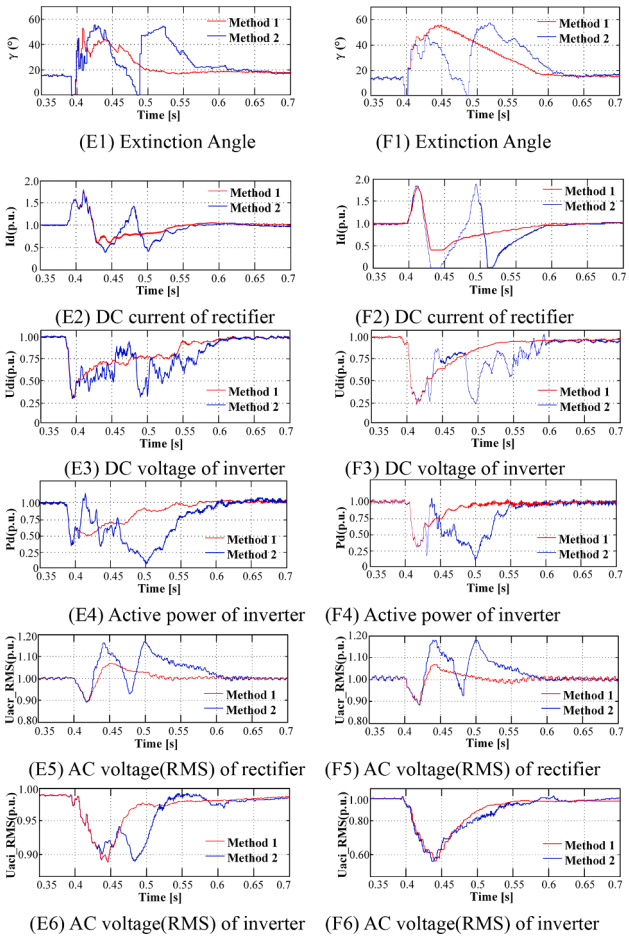


Fig. 9. Electrical quantities of LCC-HVDC under different fault depth.

LCC-HVDC control and protection system has been used. The simulation platform mainly consists of LCC-HVDC control and protection system, real time digital simulation system (RTDS), and interface device.

The control and protection system is used to realize the control and protection functions of LCC-HVDC. RTDS is used to realize the simulation of LCC-HVDC. The control system includes the proposed adaptive DC current control. The input signals of control system are provided by RTDS, including EA, DC voltage, DC current, AC voltage, AC current, etc. The control system includes VDCOL, CEAC, CCC, and CEC. The output of control system is provided to RTDS. The proposed adaptive DC current control is implemented in RTDS. The exchange of signals between the two systems is done through the interface device.

A HVDC project in China is simulated in this testing. The performance of the following two control methods is compared: Method 1 is the adaptive current control method proposed in this paper. Method 2 is the VDCOL control. The effect of Method 1 is tested under different fault depth. The three-phase fault occurs at the AC bus of the inverter station at 0.4 s. The AC bus voltage of the inverter station drops to 0.89p.u. and 0.58p.u. respectively after the fault. The experimental results captured during real-time simulation are presented in Fig. 9. The electrical quantities when using Method 1 and Method 2 are shown as the red and blue curves in Fig. 9, respectively. As can be seen from Fig. 9, the results obtained from the experiments match the software simulation results shown in Fig. 8. Compared with Method 2, Method 1, that is, the proposed method can effectively suppress the overvoltage at the send-end grid caused by CF and prevent the occurrence of SCF.

7. Conclusion

The existing control system is difficult to meet the demand for CF recovery of the inverter station and overvoltage suppression at the sending-end grid simultaneously due to the mutual influence of the electrical quantities of the inverter station and the rectifier station. Thus, a special cascading reaction is produced, and threatens the safe and stable operation of the power grid. The feasible range of DC current to avoid the high voltage ride through of wind generators at the sending-end grid and SCF of the inverter station is deduced. The feasible power region for suppressing the CF caused cascading reaction is constructed. The results demonstrate the proposed method can effectively avoid the subsequent CF and inhibit the overvoltage at the sending-end grid compared with VDCOL. Through the DC current control of the proposed method, the coordination of exchanged reactive power of rectifier station, active power and exchanged reactive power of inverter station is realized under the condition of interrupting the CF caused cascading reaction. The proposed method effectively avoids the CF caused cascading reaction in wind power transmission system via LCC-HVDC and is conducive to improving the stability of the power grid under fault.

CRediT authorship contribution statement

Mingyu Pang: Investigation, Validation, Formal analysis, Visualization, Software, Writing – original draft. **Jinxin Ouyang:** Conceptualization, Methodology, Software, Supervision. **Jianfeng Yu:** Investigation, Writing – review & editing. **Jiyu Chen:** Writing – review & editing. **Junjun Ye:** Data curation, Writing – review & editing. **Yanbo Diao:** Writing – review & editing. **Chao Xiao:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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